

KARTHIK NARRAVULA

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TECHNICAL SKILLS

Languages: Java, C++, Python

Backend: Spring Boot, Spring Security, Spring Data JPA, REST APIs, WebSocket, STOMP, Hibernate, JWT, OAuth2, Maven

Databases: SQL, PostgreSQL, Redis, MongoDB

Cloud & DevOps: AWS (EC2, RDS, S3), Docker, Nginx, Git, GitHub Actions, CI/CD

Testing: JUnit, Mockito, Integration Testing

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

PROJECTS

ConnectChat – Real-Time Chat Platform *Java · Spring Boot · WebSocket · STOMP · PostgreSQL · Docker · AWS · OAuth2*

[Live Demo](#) · [GitHub repo](#)

- Built persistent WebSocket + STOMP messaging to replace HTTP polling, and tested it to **1,500–2,500** concurrent connections with **99.9%** handshake success and **2,000–4,000** private messages/sec, holding connection drops under **0.1%** on AWS EC2 + RDS.
- Benchmarked end-to-end message delivery at **20–80ms** average (**p95: 50–100ms**, **p99: 80–150ms**) with **99.9%+** delivery success, and tracked online presence in real time via WebSocket session lifecycle events.
- Load-tested REST APIs to **3,000–6,500 req/sec** at **0% error rate** across login, search, chat history, and presence endpoints, secured via JWT access/refresh token rotation, Spring Security filter chains, and Google OAuth2 for stateless multi-path login.
- Optimized database queries by removing **N+1** patterns and adding indexes to a normalized PostgreSQL schema, keeping most endpoints under **40ms p95** while the sort- and join-heavy conversation list endpoint sustained **200–500 req/sec** at a **300–600ms p95**.
- **Deployed** the backend to a live HTTPS endpoint in under an hour by containerizing it with Docker and setting up a GitHub Actions pipeline that automatically builds, pushes to Docker Hub, and redeploys on EC2 on every push to main.
- Documented every endpoint via Swagger/OpenAPI so anyone can test the API at api.karthiknarravula.dev/swagger-ui/index.html.

knurls.me – URL Shortener *Java · Spring Boot · PostgreSQL · Redis · React · JWT · OAuth2 · AWS · Nginx*

[knurls.me](#) · [GitHub repo](#)

- Built a full URL shortening service from scratch — custom aliases, configurable TTL, and click analytics — and shipped it to [knurls.me](#) without any third-party shortening library.
- Tested authentication endpoints under **50,000 requests** with **100% success** and **0 failures**: login averaged **118ms** (p95: **205ms**) at **185 req/sec**, register averaged **185ms** at **95 req/sec**, and refresh/logout stayed under **32ms** average.
- Cached short-code lookups in Redis with a **24-hour** TTL, cutting redirect latency from **28ms** average on database hits to **4ms** average (p95: **7ms**) on cache hits, with a **96%** cache hit ratio and throughput up to **2,300 req/sec**.
- Protected the service against abuse by building a Redis-backed rate limiter using atomic increments, blocking any IP that exceeds **50 requests per minute**.
- Built a per-link analytics dashboard tracking browser, device type, referrer, and timestamp for every click, backed by indexed PostgreSQL queries averaging **92ms** (p95: **145ms**) at **170 req/sec**.
- Added Google OAuth2 alongside JWT + BCrypt so users can sign in with Google or a password — both paths share the same stateless, role-based (USER / ADMIN) security model.
- Hardened the deployment by isolating PostgreSQL and Redis inside a private AWS VPC with Nginx handling TLS termination, keeping the app stable under load — CPU averaged **58%** (peak **81%**), JVM GC pauses averaged **5ms**, and the connection pool never exceeded **16 of 20** connections.

Patient Service System – Spring Boot · Kafka · gRPC · Eureka · OpenFeign · Spring Cloud Gateway [GitHub repo](#)

- Explored microservices architecture by decomposing patient management workflows into independently deployable Spring Boot services.
- Implemented service discovery using Eureka and centralized request routing with Spring Cloud Gateway.
- Integrated gRPC and OpenFeign for inter-service communication and experimented with Kafka-based asynchronous event delivery.

EDUCATION

Takshashila University

B.Tech in Computer Science & Engineering (OOP, DBMS, OS, CN, COMPILER DESIGN)

2023 – 2027